

# SkyMap Dipole Sweep

## Results, Context, and HHT/Chirp Integration

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### 1. Executive Summary

We ran SkyMap dipole sweeps on the BAO ratio field  $R_{\text{BAO}} = \text{DH\_over\_rd} / \text{DM\_over\_rd}$  at NSIDE=32 with FWHM=6° smoothing, for Galactic latitude masks  $|b| \geq \{0,10,20,30\}^\circ$ . The catalog does not include plain DH/DM, so both the “FR-safe” sweep and the “raw” sweep used the ratio inputs.

Across masks, the measured dipole amplitudes  $A_{\text{eff}}$  were  $\sim 1.0\text{--}1.2 \times 10^{-3}$ , with permutation p-values in the  $\sim 0.036\text{--}0.263$  range for the specific examples below. The best axis-closest case (minimum  $\Delta$  to the FR reference axis) occurred at  $|b| \geq 30^\circ$  ( $\Delta \approx 60.55^\circ$ ).

### 2. Data and Configuration

- Catalog: /Users/boyde/Downloads/unified\_catalog\_REAL\_MIN.csv (missing DH/DM; contains DH\_over\_rd and DM\_over\_rd).
- Mode:  $R_{\text{BAO}} = \text{DH\_over\_rd} / \text{DM\_over\_rd}$ .
- Smoothing: NSIDE=32, FWHM=6°.
- Permutations: 1000 per map for p-value estimation ( $p \equiv \text{Pr}(A_{\text{null}} \geq A_{\text{obs}})$ ).
- Reference axes for deltas: FR (l,b)=(264°, +48°), KdS (l,b)=(300°, +30°).
- Runs created both JSON summaries and logs; the reporter aggregates both (JSON preferred for  $\Delta_{\text{FR}}, \Delta_{\text{KdS}}$ ).

### 3. Results: sweeps\_real (FR mode)

| b <br>mas<br>k (°) | FWH<br>M (°) | A_eff           | l (°)          | b (°)          | p_perm       | $\Delta_{\text{FR}}$ (°) | $\Delta_{\text{KdS}}$ (°) | axis<br>$\Delta_{\text{min}}$ (°) |
|--------------------|--------------|-----------------|----------------|----------------|--------------|--------------------------|---------------------------|-----------------------------------|
| 0                  | 6            | 0.0011557<br>40 | 151.6009<br>34 | -62.9638<br>41 | 0.03596<br>4 | 38.93733<br>7            | 38.68056<br>5             | 75.83642<br>9                     |
| 10                 | 6            | 0.0011226<br>10 | 108.1596<br>11 | -73.9254<br>08 | 0.09490<br>5 | 27.97704<br>9            | 44.34520<br>4             | 88.68184<br>1                     |
| 20                 | 6            | 0.0011627<br>70 | 76.54844<br>0  | -69.8673<br>44 | 0.13586<br>4 | 22.16462<br>7            | 46.69804<br>4             | 81.94138<br>0                     |
| 30                 | 6            | 0.0009731<br>77 | 168.5277<br>68 | -31.5377<br>45 | 0.26273<br>7 | 63.69848<br>6            | 41.37943<br>0             | 60.54841<br>8                     |

Notes: These match your recent FR sweep outputs (report\_FR.csv).

#### 4. Results: sweeps\_raw (with --fr-compatible rerun)

After re-running the raw sweep with --fr-compatible, the numerical dipole estimates matched the FR set because both ultimately used the same ratio inputs. The reporter may list duplicates when both JSON and logs exist; JSON rows include 'fr\_compatible=True', while log-derived rows default to False.

| b  mask<br>(°) | FWHM (°) | A_eff       | p_perm   | rbao_mode             | fr_compatible                |
|----------------|----------|-------------|----------|-----------------------|------------------------------|
| 0              | 6        | 0.001155740 | 0.035964 | DH_over_rd/DM_over_rd | True/False<br>(both present) |
| 10             | 6        | 0.001122610 | 0.094905 | DH_over_rd/DM_over_rd | True/False<br>(both present) |
| 20             | 6        | 0.001162770 | 0.135864 | DH_over_rd/DM_over_rd | True/False<br>(both present) |
| 30             | 6        | 0.000973177 | 0.262737 | DH_over_rd/DM_over_rd | True/False<br>(both present) |

Deduping tip: keep one row per (mask,fwhm); prefer JSON over logs to retain  $\Delta_{\text{FR}}/\Delta_{\text{KdS}}$ .

#### 5. Interpretation & Implications

- R\_BAO dipole amplitude  $\sim 10^{-3}$  across masks.
- Axis varies with mask; closest to FR axis at  $|b| \geq 30^\circ$  ( $\approx 60.55^\circ$ ).
- p-values  $\sim 0.036$ – $0.263$  here; increase permutations and vary NSIDE/FWHM to refine significance.
- No numerical difference between 'FR' and 'raw' runs because both used the ratio inputs in this catalog.

#### 6. Where HHT & Chirp Structures Fit

Context: HHT on SN residuals shows IMF2/IMF3 features and IMF2 'locking' to BAO-relevant scales; tested via robustness grids and surrogate analysis.

Connection to R\_BAO dipole: Anisotropic coupling from BAO-linked IMF2 energy could project to low- $\ell$  dipole. Chirp vs  $\log(1+z)$  suggests scale drift; mask-dependent axis shifts are compatible with footprint/selection effects. Combine map-level dipole tests with HHT ridge/surrogate significance for a joint detection framework.

Operationally: use per-mask CSV snapshots; recompute HHT ridge metrics within each mask's footprint; sweep NSIDE/FWHM alongside HHT grids.

## 7. Relation to the PLamb Theory Document

The PLamb document (FR/KdS focus, accretion-driven acceleration  $A_{\text{acc}}$ ) connects parameter-level inferences to sky-level anisotropy tests. Current working fit (subject to refinement):  $H_0 \approx 71.25$ ,  $\Omega_m \approx 0.279$ ,  $\Omega_\Lambda \approx 0.739$ ,  $A_{\text{acc}} \approx 0.030$ . We use FR/KdS reference axes to compute  $\Delta_{\text{FR}}$  and  $\Delta_{\text{KdS}}$ , tying the observed  $R_{\text{BAO}}$  dipole to theory.

## 8. Note on CMB ‘Ripples’ from Bubble Collisions (Hypothesis)

A lecture suggested possible large-scale CMB ripples from multiverse bubble collisions. Interesting but speculative here; if pursued, cross-check axis alignments and low- $\ell$  patterns against the  $R_{\text{BAO}}$  dipole with robust null tests.

## 9. Recommended Next Steps

- Obtain a catalog with true DH and DM to compare against the ratio mode directly.
- Increase permutations (e.g., 10k) and sweep  $\text{NSIDE} \in \{16, 32, 64\}$ ,  $\text{FWHM} \in \{4^\circ, 6^\circ, 8^\circ\}$ .
- Add deduping (prefer JSON over logs) in the reporter to avoid duplicates.
- Build joint panels per mask:  $\{A_{\text{eff}}, \Delta_{\text{FR}}, \Delta_{\text{KdS}}, p\} \leftrightarrow \{\text{IMF2 energy, chirp slope, surrogate } p\}$ .
- If CMB hypothesis pursued, design a falsifiable cross-correlation test.

## Appendix A — Reproducibility

```
%run "/Users/boyde/.spyder-py3/sm_quickgrid.py" --catalog  
"/Users/boyde/Downloads/unified_catalog_REAL_MIN.csv" --rbao-mode DH_over_rd/DM_over_rd  
--fr-compatible --nside 32 --fwhm-smooth-deg 6 --gal-mask-list 0,10,20,30 --use-gal-mask-in-  
analysis --use-gal-mask-in-display --perm-n 1000 --outdir  
"/Users/boyde/.spyder-py3/plamb_runs/SkyMap/figs/sweeps_real"
```

```
%run "/Users/boyde/.spyder-py3/sm_sweep_report.py" --dir  
"/Users/boyde/.spyder-py3/plamb_runs/SkyMap/figs/sweeps_real" --out "/Users/boyde/.spyder-  
py3/plamb_runs/SkyMap/figs/sweeps_real/report_FR" --truth-A 0.0010 --truth-axis "230,-10"
```

```
%run "/Users/boyde/.spyder-py3/sm_quickgrid.py" --catalog  
"/Users/boyde/Downloads/unified_catalog_REAL_MIN.csv" --rbao-mode DH/DM --fr-compatible --  
nside 32 --fwhm-smooth-deg 6 --gal-mask-list 0,10,20,30 --use-gal-mask-in-analysis --use-gal-mask-  
in-display --perm-n 1000 --outdir  
"/Users/boyde/.spyder-py3/plamb_runs/SkyMap/figs/sweeps_raw"
```

```
%run "/Users/boyde/.spyder-py3/sm_sweep_report.py" --dir  
"/Users/boyde/.spyder-py3/plamb_runs/SkyMap/figs/sweeps_raw" --out "/Users/boyde/.spyder-  
py3/plamb_runs/SkyMap/figs/sweeps_raw/report_raw" --truth-A 0.0010 --truth-axis "230,-10"
```